

Krishin Lal

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Professional Summary

AI Engineer & Data Scientist with hands-on production experience building LLM powered systems, agentic pipelines, OCR automation, geospatial analytics, and deep learning forecasting at one of Pakistan's largest banks. Specializes in end-to-end AI system design from embedding based entity matching and vector retrieval to FastAPI deployment on Azure ML with a consistent track record of replacing manual workflows with scalable, measurable AI solutions.

Experience

Data Scientist, United Bank Limited (UBL) - Karachi, Pakistan July 2025 – Present

- **Procurement Invoice OCR System:** Engineered a production ready multi-class invoice processing pipeline (Class A, B, C) using Azure Document Intelligence, GPT-4o, and FastAPI, cutting per invoice processing time from 5 to 1 minute (80% reduction). Across 150 invoices/day, this delivers 10 hours of daily savings and 200+ hours reclaimed monthly, freeing the finance team from manual data entry entirely.
- **Trade Document OCR System:** Replaced a 5-hour manual daily process with a 15-minute automated solution using Python, Azure Document Intelligence, and a custom LLM-based validation layer capable of handling handwritten content in UBL trade batch cover pages achieving 1,467 annual hours saved (from 1,500 to 33 hours/year).
- **Enterprise Account Opening Automation:** Developed end-to-end OCR pipeline processing account opening forms across all UBL branches nationwide, extracting structured and unstructured fields (KYC, customer info, signatures) and outputting validated JSON for direct system integration eliminating a costly third-party vendor dependency and establishing a reusable in-house document automation framework.
- **NTB/ETB Entity Matching Engine:** Designed a semantic matching algorithm using SentenceTransformers (all-MiniLM-L6-v2) and cosine similarity to reconcile thousands of company names from Apollo CRM against UBL's bank account records, classifying them as Existing-to-Bank (ETB) or New-to-Bank (NTB) prospects. Implemented blocking indexing, compact key normalization, domain noise filtering, and a multi-stage match pipeline (exact, containment, embedding) with configurable confidence thresholds enabling the business development team to identify verified NTB prospects for targeted acquisition outreach.
- **Geospatial Branch Expansion Planning (ABEP):** Engineered end-to-end geospatial intelligence system for UBL's Annual Branch Expansion Plan. Mapped coordinates for 2,200+ UBL and newly acquired Silk Bank branches using SERP API. Extracted 8,000+ competitor branch addresses from HBL, ABL, Meezan, MCB, NBP, BAFL, and BAHF across KLRIM (Karachi, Lahore, Islamabad, Rawalpindi, Multan) via developer tools and geocoded them at scale. Constructed 1x1 km geospatial grids over Pakistan's highest-revenue cities, overlaid all branch locations, and identified competitor-dense blind spots with no UBL presence. Applied a 500m cannibalization buffer around existing UBL branches to carve out viable opportunity polygons producing 116 data-backed branch opportunity zones and directly influencing executive decisions on branch opening strategy.
- **Branch Cash Optimization:** Built hourly-granularity cash demand forecasting system for UBL branches using the NeuralForecast framework with N-BEATS, benchmarked against SARIMA, ARIMA, and DeepAR. Processed branch-level transaction data filtered by cash category and business hours (weekdays, 9–17), trained per-branch models with walk-forward validation, and evaluated using RMSE and a within-bounds accuracy metric ($\pm 25\%$ tolerance band) achieving $\sim 85\%$ accuracy on held-out data enabling branches to right-size daily cash reserves and eliminate excessive vault holdings.
- **Production ML Infrastructure:** Containerized and deployed 5-6 production models on Azure Machine Learning with FastAPI inference endpoints, authentication, request validation, and monitoring dashboards collectively processing 600-700 daily inference requests per endpoint.

Projects

Multi-Agent Narrative System | Hackfest x Datathon – Generative AI Competition 2024

- Built a multi-agent AI system using LangGraph and Google Gemini featuring a Director agent that orchestrates narrative flow across multiple character agents, each with persistent memory and a distinct personality profile.

- Implemented a non-verbal action system enabling agents to express physical actions and emotional states alongside dialogue, significantly improving narrative coherence in AI-generated storytelling.
- Designed inter-agent communication and memory-sharing protocols so characters retain context across long narrative arcs, demonstrating practical application of stateful agentic architectures.

EDNOVA – Gamified SaaS Learning Management System (Final Year Project)

Mar 2024 – Jun 2025

- Architected and shipped a full-stack gamified LMS using the MERN stack, featuring real-time engagement tracking (study duration, video watch time, quiz participation), a points-based reward engine, and dynamic leaderboards.
- Built a granular analytics dashboard enabling educators to identify at-risk students early by surfacing composite engagement scores across assignments, quizzes, and lecture materials.
- Designed a scalable multi-tenant SaaS architecture with role-based access control (students, instructors, admins), RESTful APIs, and a responsive frontend supporting concurrent users.

Skills

AI & LLM Engineering: LangChain, LangGraph, Large Language Models (LLMs), RAG Architecture, Agentic Systems, Multi-Agent Orchestration, Prompt Engineering, Semantic Search, Vector Embeddings, Azure OpenAI

Computer Vision & OCR: Azure Document Intelligence, Document Layout Analysis, Table Extraction, Handwriting Recognition, Image Preprocessing

Machine Learning & Deep Learning: Scikit-learn, TensorFlow, Keras, XGBoost, N-BEATS, NHITS, SARIMA, ARIMA, DeepAR, LSTM, Prophet, NeuralForecast, Time-Series Forecasting, Classification, Regression, Hyperparameter Tuning

NLP & Embeddings: SentenceTransformers, HuggingFace, Cosine Similarity, Entity Matching, Semantic Blocking, ChromaDB, Vector Databases

Geospatial Engineering: SERP API, Coordinate Extraction, Grid-based Spatial Analysis, Geospatial Blind Spot Detection, Cannibalization Radius Modeling, Opportunity Zone Mapping

Deployment & MLOps: FastAPI, Azure Machine Learning, Docker, Model Containerization, REST API Design, API Authentication, Request Validation, Endpoint Monitoring

Data Engineering & Databases: Pandas, NumPy, SQL, Teradata, MongoDB, SQL Server, Feature Engineering, EDA

Development Tools & Platforms: Python, SQL, JavaScript (MERN Stack), Git, GitHub, Docker, Jupyter, VS Code, Azure Cloud Services, Kubeflow,

Education

Iqra University, Karachi – Bachelor of Science in Computer Science

Oct 2021 – Jun 2025

Government Degree College, Kandhkot – Intermediate (Science)

2019 – 2021

Key Achievements

- **1,467 hours/year saved** via trade document OCR automation at UBL (1,500 → 33 hours/year).
- **10 hours/day (200+ hours/month) reclaimed** through procurement invoice automation 4 minutes saved per invoice across 150 daily vendor invoices.
- **116 geospatial branch opportunity zones** identified across Pakistan, directly shaping UBL's Annual Branch Expansion Plan using 1×1 km grid analysis over 8,000+ competitor locations.
- **2,479 companies** classified as NTB/ETB prospects via semantic embedding matching engine, enabling targeted business development outreach.
- **600–700 daily inference requests** served across production ML endpoints deployed on Azure Machine Learning.
- **Eliminated third-party vendor dependency** by building in-house account opening OCR automation.